

Chen-Chieh Liao

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Research Interests

Human-computer interaction, machine learning, 3D vision, virtual and augmented reality, human motion analysis, and intelligent systems for motor-skill acquisition.

Appointments

Institute of Science Tokyo Assistant Professor, School of Computing	Jan. 2026–Present
The University of Tokyo Guest Researcher, Faculty of Medicine	Jun. 2025–Present
Institute of Science Tokyo Postdoctoral Researcher	Apr. 2025–Dec. 2025
Cygames, Inc. Associate Researcher	Jan. 2024–Aug. 2025
Keio University Visiting Researcher, Graduate School of Media Design	Apr. 2023–Mar. 2025
NAVER Cloud AI Researcher	Sep. 2022–Mar. 2023
ACES, Inc. Algorithm Engineer	Jul. 2022–Feb. 2024

Education

Institute of Science Tokyo Ph.D. in Computer Science. Advisor: Prof. Hideki Koike.	Apr. 2022–Mar. 2025
Tokyo Institute of Technology M.S. in Computer Science. Advisor: Prof. Hideki Koike.	Apr. 2020–Mar. 2022
Tokyo Institute of Technology B.E. in Computer Science. Overall average: 95.33/100; major average: 96.56/100.	Apr. 2016–Mar. 2020
ETH Zurich Exchange Program, Department of Computer Science.	Sep. 2018–Aug. 2019

Grants, Fellowships, and Scholarships

Japan Society for the Promotion of Science Grant-in-Aid for Early-Career Scientists.	Apr. 2026
Japan Society for the Promotion of Science Research Fellowship for Young Scientists (DC2).	Apr. 2023
MEXT Japanese Government Scholarship.	Apr. 2022
Moritani Scholarship Foundation Privately funded scholarship.	Apr. 2017, Apr. 2020
Heyning-Roelli Foundation Privately funded scholarship.	Sep. 2018
Tokyo Institute of Technology Fund University-funded scholarship.	Sep. 2018
Monbukagakusho Honors Scholarship Scholarship award.	Apr. 2015, Apr. 2016

Teaching Experience

- Teaching Assistant, Data Structures and Algorithms, Dec. 2019–Feb. 2020.
- Teaching Assistant, Information Literacy I & II, Apr. 2020–Aug. 2020 and Sep. 2020–Feb. 2021.
- Teaching Assistant, Computer Science I & II, Apr. 2021–Aug. 2021, Sep. 2021–Feb. 2022, and Apr. 2022–Aug. 2022.

Publications

Complete and current list: [Google Scholar](#). Equal-contribution authors are marked with *.

1. C.-C. Liao, Y. Peng, Y. Cai, Yûi Ono, H. Hanaoka, E. Wu, H. Koike, and S. Kurabayashi. “Motion Style Slider: Endpoint-Supervised Continuous Style Control for Human Motion Diffusion.” *ECCV*, 2026.
2. Y. Peng*, J.-T. Song*, C.-C. Liao*, K. Kitani, H. Koike, and E. Wu. “InterPet4D: A Multimodal 4D Human–Pet Interaction Dataset for Pet Motion Generation.” *ECCV*, 2026.
3. R. Liu, C.-C. Liao, T. Takahashi, Y. Peng, E. Wu, and H. Koike. “MuscleGolfAR: Embodied versus Detached Visualization for Motions and Inferred Muscle Activations in Augmented Reality.” *IEEE TVCG*, 32(5):3093–3103, 2026. [DOI](#)
4. T. Hirano, H. Yoshihara, Y. Peng, C.-C. Liao, E. Wu, and H. Koike. “SoleCoach: Sole Pressure and IMU-based MLLMs for Skill Coaching.” *CHI*, 2026. [DOI](#)
5. K. Chen, R. Panskus, E. Wu, Y. Peng, D. Saito, E. Kamiyama, R. Li, C.-C. Liao, K. Marky, A. Kato, H. Koike, and K. Kunze. “Sensing Your Vocals: Exploring the Activity of Vocal Cord Muscles for Pitch Assessment Using Electromyography and Ultrasonography.” *CHI*, 2026. [DOI](#)
6. H. Yoshihara, T. Hirano, C.-C. Liao, E. Wu, and H. Koike. “Real-time Ski Coaching using Pose Estimation from SolePressure and MLLMs Based on Reference-Pose Errors.” *Augmented Humans*, 2026. [DOI](#)
7. S. Watanabe, C.-C. Liao, E. Wu, and H. Koike. “Seeing and Saying Movement: Combining Skeleton Visualization and LLM Feedback for Motor Skill Coaching.” *Augmented Humans*, 2026. [DOI](#)
8. Y. Tabei, C.-C. Liao, Y. Peng, E. Wu, and H. Koike. “An AR-Based Exergame for Supporting Posture Correction in Alpine Skiing.” *IEEE VR Workshops*, 2026. [DOI](#)
9. S. Ito, Y. Peng, C.-C. Liao, S. Watanabe, E. Wu, and H. Koike. “Exploring Foot Pressure Sensing for Avatar Manipulation and for Input Modality in VR.” *IEEE VR Workshops*, 2026. [DOI](#)
10. F. Oba, Y. Peng, C.-C. Liao, E. Wu, N. Yoshimura, and H. Koike. “ElectroEncephaloGraphy Verification of Speed Adaptation in Time-Distortion Ski Training System.” *IEEE VR Workshops*, 2026. [DOI](#)
11. R. Liu, Y. Peng, T. Oku, C.-C. Liao, E. Wu, S. Furuya, and H. Koike. “From Pose to Muscle: Multimodal Learning for Piano Hand Muscle Electromyography.” *NeurIPS*, 2025.
12. C.-C. Liao, R. Liu, Y. Peng, E. Wu, and H. Koike. “A Demonstration of ShiftingGolf: Gross Motor Skill Correction Using Redirection in VR.” *SIGGRAPH Asia Emerging Technologies*, 2025. [DOI](#)
13. Y. Tabei, C.-C. Liao, Y. Peng, E. Wu, and H. Koike. “An Augmented Reality Exergame for Improving Ski Posture through Gamified Feedback.” *SIGGRAPH Asia Emerging Technologies*, 2025. [DOI](#)
14. K. Chen, E. Wu, C.-C. Liao, D. Saito, Y. Peng, A. Kato, H. Koike, and K. Kunze. “Introduction of the VCSD Dataset: A Vocal Cords Dataset Using EMG and Ultrasonography for Singing Pitch Skill Recognition.” *UbiComp Companion*, 2025. [DOI](#)
15. T. Hirano, Y. Tabei, Y. Peng, C.-C. Liao, E. Wu, and H. Koike. “SkiTechCoach: A Multimodal Alpine Skiing Dataset with 3D Body Pose, Sole Pressure, and Expert Coaching.” *MMSports*, 2025. [DOI](#)
16. R. Liu, Y. Peng, T. Oku, C.-C. Liao, E. Wu, S. Furuya, and H. Koike. “PiaMuscle: Improving Piano Skill Acquisition by Cost-effectively Estimating and Visualizing Activities of Miniature Hand Muscles.” *CHI*, 2025. [DOI](#)
17. C.-C. Liao, Z. Yu, and H. Koike. “ShiftingGolf: Gross Motor Skill Correction Using Redirection in VR.” *IEEE TVCG*, 31(5):3429–3439, 2025. [DOI](#)
18. K. Hayakawa, D.-H. Hwang, C.-C. Liao, and H. Koike. “Diffusion-Based Synthetic Dataset Generation for Egocentric 3D Human Pose Estimation.” *ECCV Workshops*, 2024. [DOI](#)
19. K. Chen, E. Wu, D. Saito, Y. Peng, C.-C. Liao, A. Kato, H. Koike, and K. Kunze. “Novel Sensing Methods for Vocal Technique Analysis: Evaluation on Electromyography and Ultrasonography.” *ISMAR Adjunct*, 2024. [DOI](#)
20. T. Matsumoto, E. Wu, C.-C. Liao, and H. Koike. “ARpenSki: Augmenting Ski Training with Direct and Indirect Postural Visualization.” *IEEE VR*, 2024. [DOI](#)
21. E. Wu, T. Matsumoto, C.-C. Liao, R. Liu, H. Katsuyama, Y. Inaba, N. Hakamada, Y. Yamamoto, Y. Ishige, and H. Koike. “SkiTech: An Alpine Skiing and Snowboarding Dataset of 3D Body Pose, Sole Pressure, and Electromyography.” *MMSports*, 2023. [DOI](#)

22. R. Liu, E. Wu, C.-C. Liao, H. Nishioka, S. Furuya, and H. Koike. “PianoSyncAR: Enhancing Piano Learning through Visualizing Synchronized Hand Pose Discrepancies in Augmented Reality.” *ISMAR*, 2023. [DOI](#)
23. C.-C. Liao, J.-H. Kim, H. Koike, and D.-H. Hwang. “Content-Preserving Motion Stylization using Variational Autoencoder.” *SIGGRAPH Posters*, 2023. [DOI](#)
24. Y. Sun, Q. Wu, H. Zhou, K. Wang, T. Hu, C.-C. Liao, S. Miyafuji, Z. Liu, and H. Koike. “Make Your Brief Stroke Real and Stereoscopic: 3D-Aware Simplified Sketch to Portrait Generation.” *ICMI*, 2023. [DOI](#)
25. R. Liu, E. Wu, C.-C. Liao, H. Nishioka, S. Furuya, and H. Koike. “PianoHandSync: An Alignment-based Hand Pose Discrepancy Visualization System for Piano Learning.” *CHI Extended Abstracts*, 2023. [DOI](#)
26. C.-C. Liao, D.-H. Hwang, E. Wu, and H. Koike. “AI Coach: A Motor Skill Training System using Motion Discrepancy Detection.” *Augmented Humans*, 2023. [DOI](#)
27. C.-C. Liao, D.-H. Hwang, and H. Koike. “AI Golf: Golf Swing Analysis Tool for Self-Training.” *IEEE Access*, 10:106286–106295, 2022. [DOI](#)
28. C.-C. Liao, H. Kikuchi, D.-H. Hwang, and H. Koike. “Virtual Club Shadow: A Real-time Projection of Golf Club Trajectory.” *ISMAR Adjunct*, 2022. [DOI](#)
29. E. Wu, C.-C. Liao, R. Liu, and H. Koike. “Context-aware Risk Degree Prediction for Smartphone Zombies.” *SIGGRAPH Posters*, 2022. [DOI](#)
30. R. Liu, E. Wu, C.-C. Liao, H. Nishioka, S. Furuya, and H. Koike. “Synchronized Hand Difference Visualization for Piano Learning.” *SIGGRAPH Posters*, 2022. [DOI](#)
31. C.-C. Liao, D.-H. Hwang, and H. Koike. “How Can I Swing Like Pro?: Golf Swing Analysis Tool for Self Training.” *SIGGRAPH Asia Posters*, 2021. [DOI](#)
32. C.-C. Liao, H. Koike, and T. Nakamura. “Realtime Center of Mass Adjustment via Weight Switching Device inside a Golf Putter.” *CHI Extended Abstracts*, 2020. [DOI](#)

Technical Skills and Languages

Programming: C/C++, Java, JavaScript, Python, MATLAB, Scheme.

Tools: PyTorch, TensorFlow, OpenCV, OpenGL, OpenVR, Unity, Arduino, Git, Docker.

Languages: Mandarin (native), English (fluent; TOEIC 970/990), Japanese (fluent; JLPT N1 178/180).